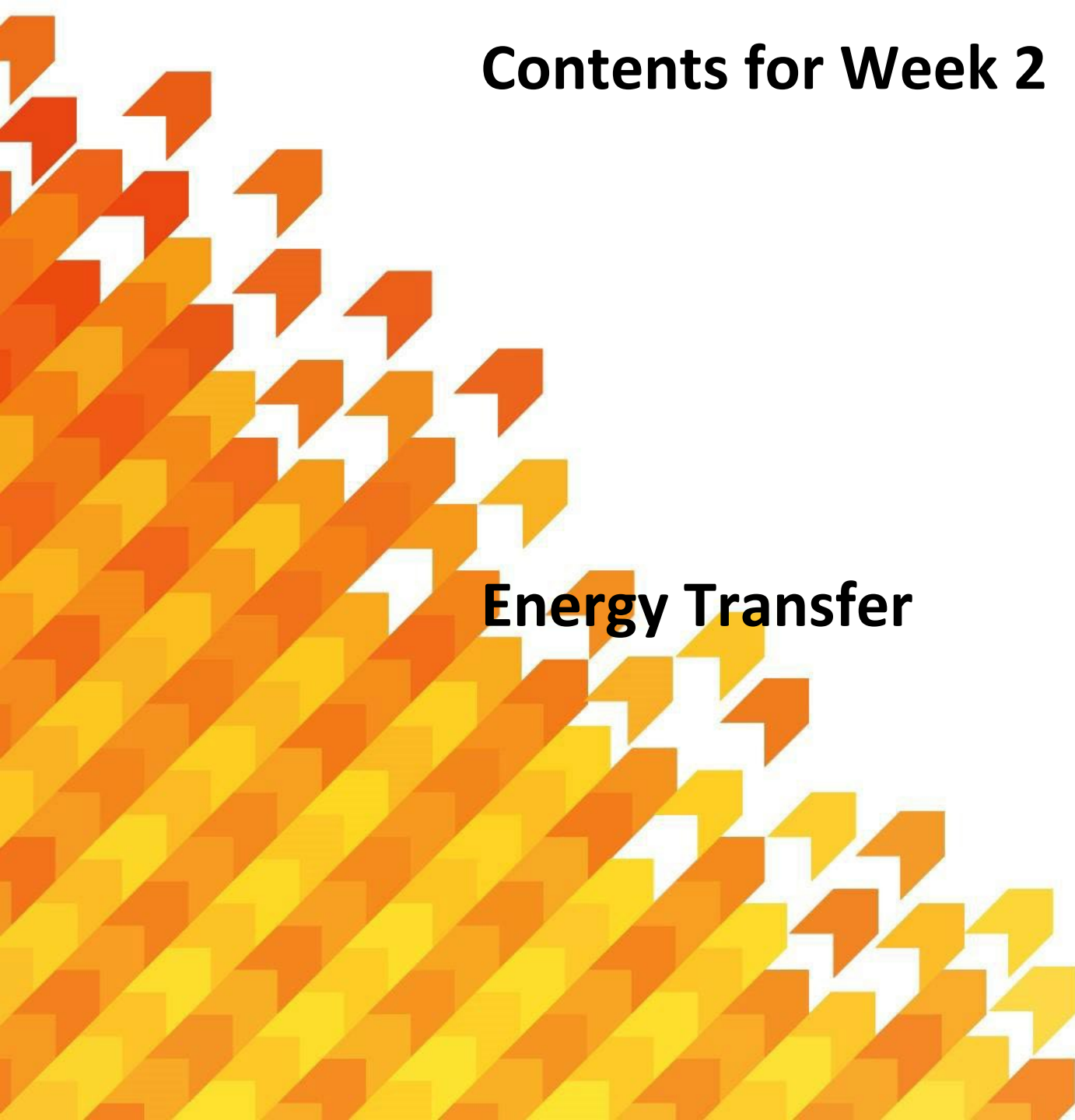


ENME601 Thermodynamics

Week 2 : Energy, Energy Transfer, and General Energy Analysis

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Contents for Week 2

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Short review - Thermodynamics and energy

Thermodynamics is the science dealing with the relations between all forms of *energy*.

Today, Thermodynamics strongly focuses on energy conversion systems (e.g. refrigerators, power plants).

Classical Thermodynamics: A *macroscopic approach* to the study of Thermodynamics.

- It involves the concept of continuum. *Large-scale properties and behaviors*.

Statistical Thermodynamics: A *microscopic approach* to the study of Thermodynamics.

- Considering the numbers of possible arrangements of *atoms and molecules within a system*, known as microscopic states.

How about the forms of energy ?

Forms of energy (1)

Energy can exist in numerous forms (e.g. thermal, mechanical, kinetic, potential, electric, chemical) and their sum constitutes the **total energy** (E) of a system. The magnetic, electric, and surface tension effects are usually ignored. In the absence of such effects:

$$E = \underbrace{U}_{\text{MICROSCOPIC FORM OF ENERGY}} + \underbrace{KE + PE}_{\text{MACROSCOPIC FORMS OF ENERGY}} = \underbrace{U}_{\text{MICROSCOPIC FORM OF ENERGY}} + m \frac{V^2}{2} + mgz \quad (\text{kJ}) \quad \text{Total energy of a system}$$
$$e = \underbrace{u}_{\text{MICROSCOPIC FORM OF ENERGY}} + \underbrace{ke + pe}_{\text{MACROSCOPIC FORMS OF ENERGY}} = \underbrace{u}_{\text{MICROSCOPIC FORM OF ENERGY}} + \frac{V^2}{2} + gz \quad (\text{kJ/kg}) \quad \text{Energy of a system per unit mass}$$

U = internal energy (u = internal energy per unit mass)

KE = kinetic energy (ke = kinetic energy per unit mass)
 PE = potential energy (pe = potential energy per unit mass)

m = mass [kg]

v = velocity of the system relative to some fixed reference frame [m/s]

g = gravitational acceleration [m/s²]

z = elevation of the center of gravity of a system relative to some arbitrarily selected reference level [m]

Thermodynamics deals only with the *change* of the total energy (ΔE) of a system, which is what matters in engineering problems.

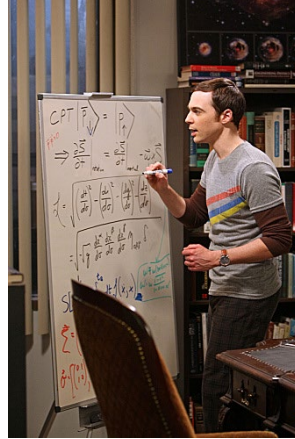
Forms of energy (2)

Internal energy is related to the *molecular structure* and the degree of *molecular activity* and can be viewed as the sum of the *kinetic* (microscopic motion of the system's particles from translations, rotations and vibrations) and *potential* (e.g. chemical bonds) energies of the molecules.

Internal energy

- **Sensible energy** = portion of the internal energy of a system associated with the kinetic energies of the molecules ($v_{\text{average molecules}}$ & molecular activities $\propto T$, i.e. $T \rightarrow > KE_{\text{molecules}} \rightarrow > U_{\text{system}}$)
- **Latent energy** = internal energy associated with the phase of a system (e.g. if enough energy is provided, the liquid phase of a system turns into gas phase)
- **Chemical energy** = internal energy associated with the atomic bonds in a molecule (e.g. during a combustion process, some chemical bonds are destroyed while others are formed $\rightarrow$ the internal energy changes)
- **Nuclear energy** = in nuclei (not usually considered in classical thermodynamics)

Forms of energy (3)



Static forms of energy = total energy of a system being contained or stored in the system



Dynamic forms of energy (or **energy interactions**) = forms of energy not stored in a system

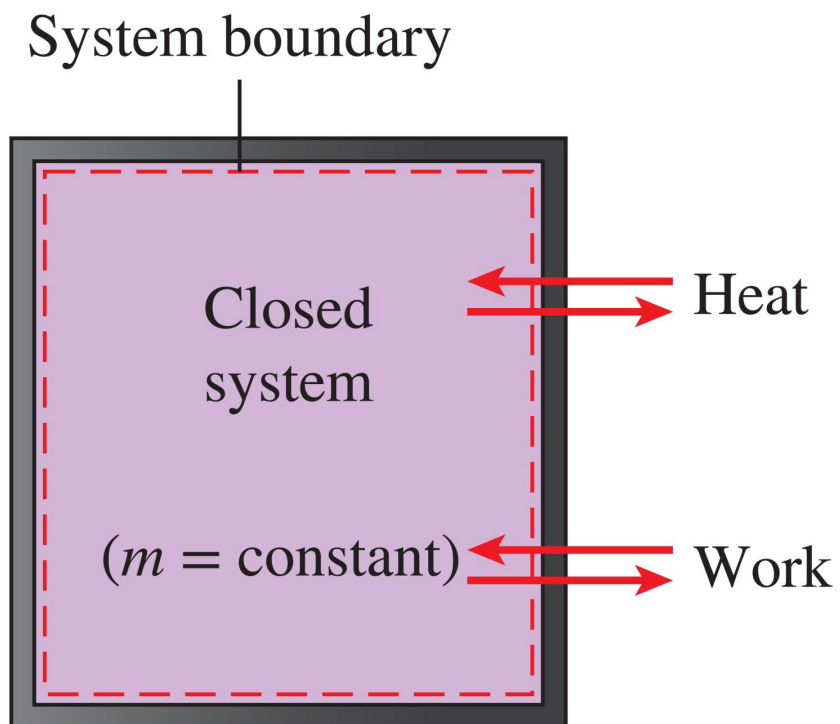
- Recognized at the system boundary as they cross it
- Heat transfer or work

Thermal = Sensible + Latent

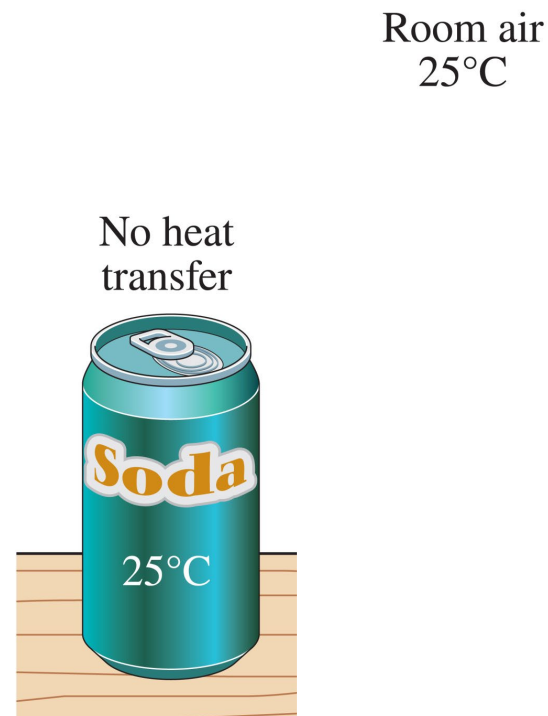
Internal = Sensible + Latent + Chemical + Nuclear

Energy transfer by heat (1)

Heat is defined as the form of energy that is transferred between two systems (or a system and its surroundings) due to a temperature difference.



Energy can cross the boundaries of a closed system in the form of heat and work.

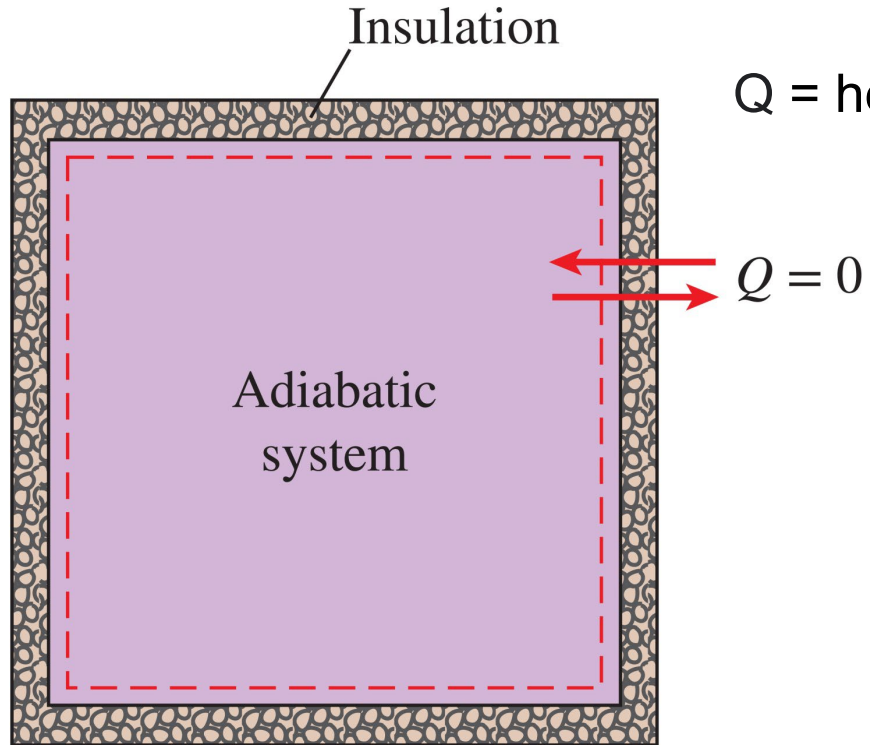


Temperature difference is the driving force for heat transfer. The larger the temperature difference, the higher is the rate of heat transfer.

$$\dot{Q} = hA (T_{\text{surroundings}} - T_{\text{object}})$$

Energy transfer by heat (2)

A process during which there is no heat transfer is called an **adiabatic process**.



Q = heat [kJ]



Energy transfer by heat (3)

The amount of heat transferred during the process between two states (states 1 and 2) is denoted by Q_{12} .

Heat transfer *per unit mass* (q_{12}) can be calculated as:

$$q_{12} = \frac{Q_{12}}{m} \left[\frac{\text{kJ}}{\text{kg}} \right]$$

q = specific heat [kJ]
 Q = heat [kJ]
 m = mass [kg]

Amount of heat transfer (Q_{12}) as *heat transfer rate* changes with time ($\dot{Q}_{12}$) can be calculated as:

overdot indicates "per unit time"

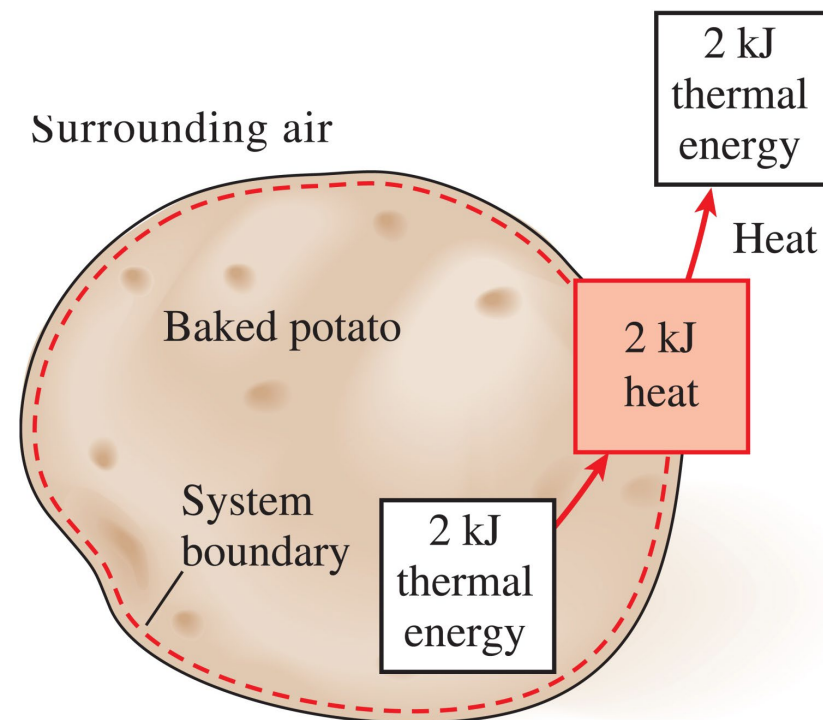
$$Q_{12} = \int_{t_1}^{t_2} \dot{Q}_{12} dt \quad [\text{kJ}]$$

Q = heat [kJ]
 $\dot{Q}$ = heat transfer rate [kW]
 dt = differential of time [s]
 t_1 = time at state 1 [s]
 t_2 = time at state 2 [s]

Amount of heat transfer (Q_{12}) as heat transfer rate ($\dot{Q}_{12}$) is constant can be calculated as:

$$Q_{12} = \dot{Q}_{12} \cdot \Delta t \quad [\text{kJ}] \quad (\Delta t = t_2 - t_1)$$

Q = heat [kJ]
 $\dot{Q}$ = heat transfer rate [kW]
 Δt = time interval [s]
 t_1 = time at state 1 [s]
 t_2 = time at state 2 [s]



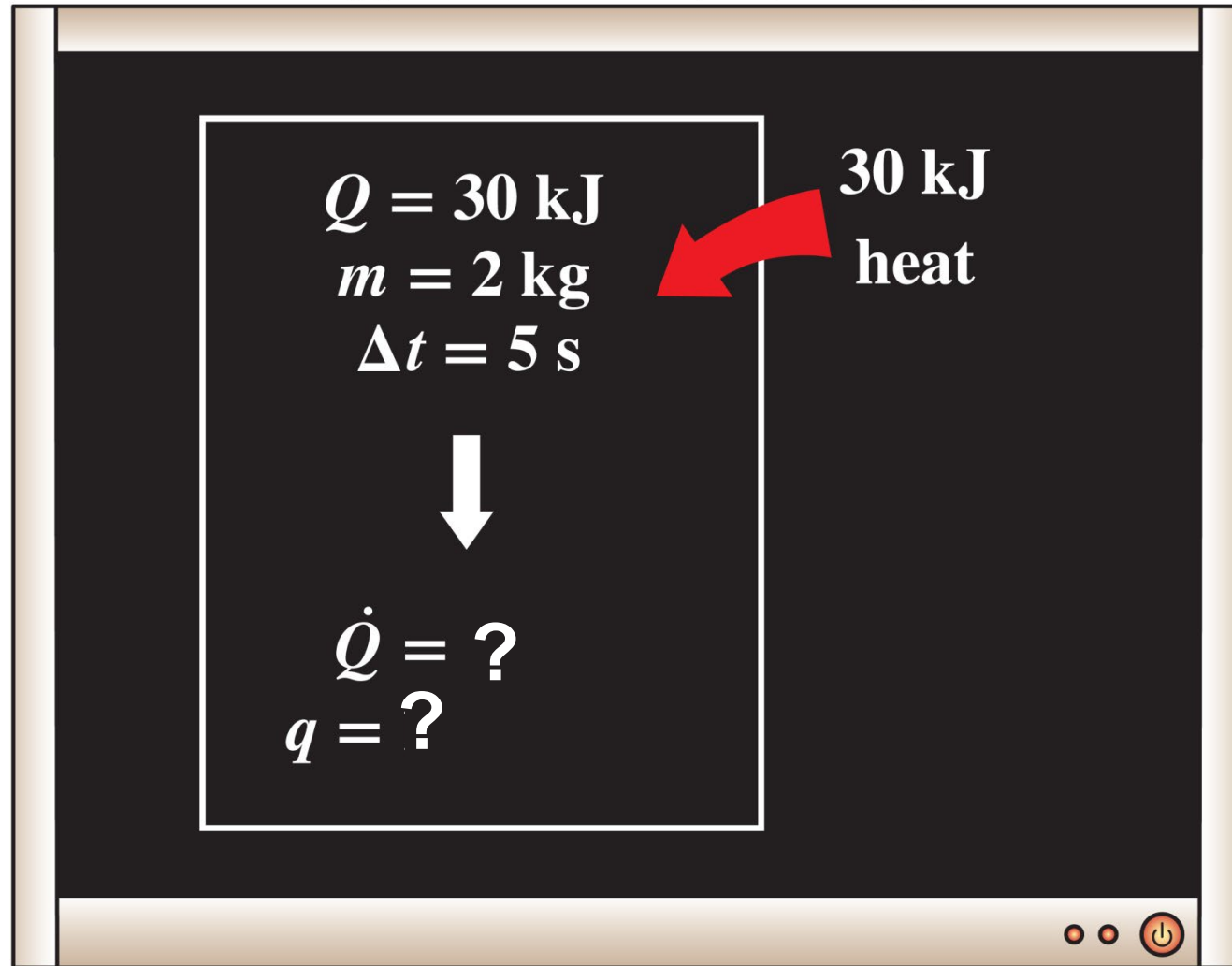
Energy is recognized as heat transfer only as it crosses the system boundary.

Short practice 1

Question: A 2 kg object receives 30 kJ of heat over a period of 5 seconds.

To find:

- 1) Calculate the heat transfer rate $\dot{Q}$ in kW
- 2) Determine the specific heat transfer q in kJ/kg.



Energy transfer by work (1)

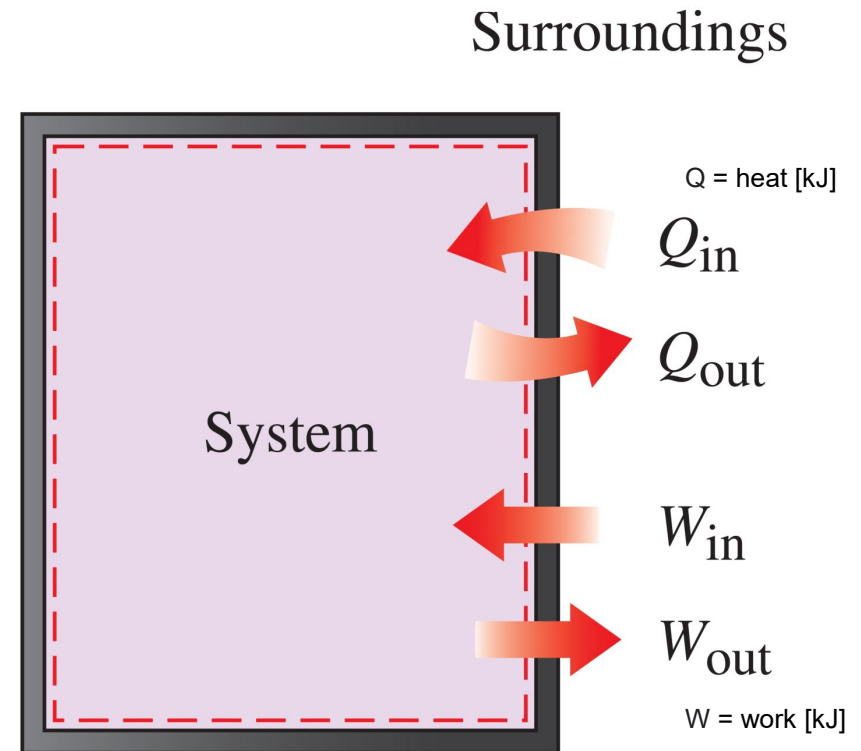
Work is the *energy transfer associated with a force acting through a distance*.

A **rising piston**, a **rotating shaft** and an **electric wire crossing the system boundaries** are all associated with work interactions.

Heat and work are *directional quantities*. The generally accepted **formal sign convention** is:

- **heat transfer to a system and work done by a system are positive;**
- **heat transfer from a system and work done on a system are negative.**

Alternative to sign convention is to use the subscripts **in** and **out** to indicate direction.



Specifying the directions
of heat and work.

Energy transfer by work (2)

Electrical work

When N coulombs of electrical charge move through a potential difference $\mathbf{V}$, the electrical work done is

$$W_e = \mathbf{V}N \quad \text{Electrical work}$$

When potential difference and current change with time

$$\dot{W}_e = \mathbf{V}I \quad (\text{W}) \quad \text{Electrical power}$$

$\dot{W}_e$ is the electrical power and I is the number of electrical charges flowing per unit time, that is, the *current*

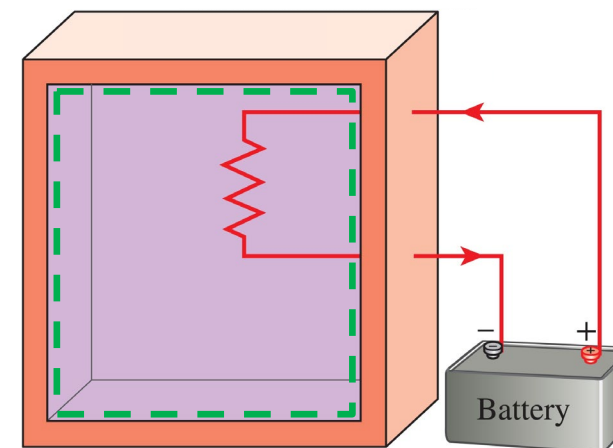
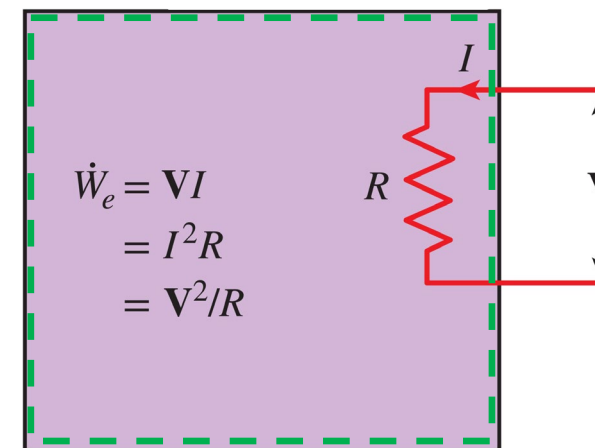
In general, both $\mathbf{V}$ and I vary with time, and the electrical work done during a time interval Δt is expressed as

$$W_e = \int_1^2 \mathbf{V}I dt \quad (\text{kJ})$$

When both $\mathbf{V}$ and I remain constant during the time interval Δt , it reduces to

$$W_e = \mathbf{V}I \Delta t \quad (\text{kJ})$$

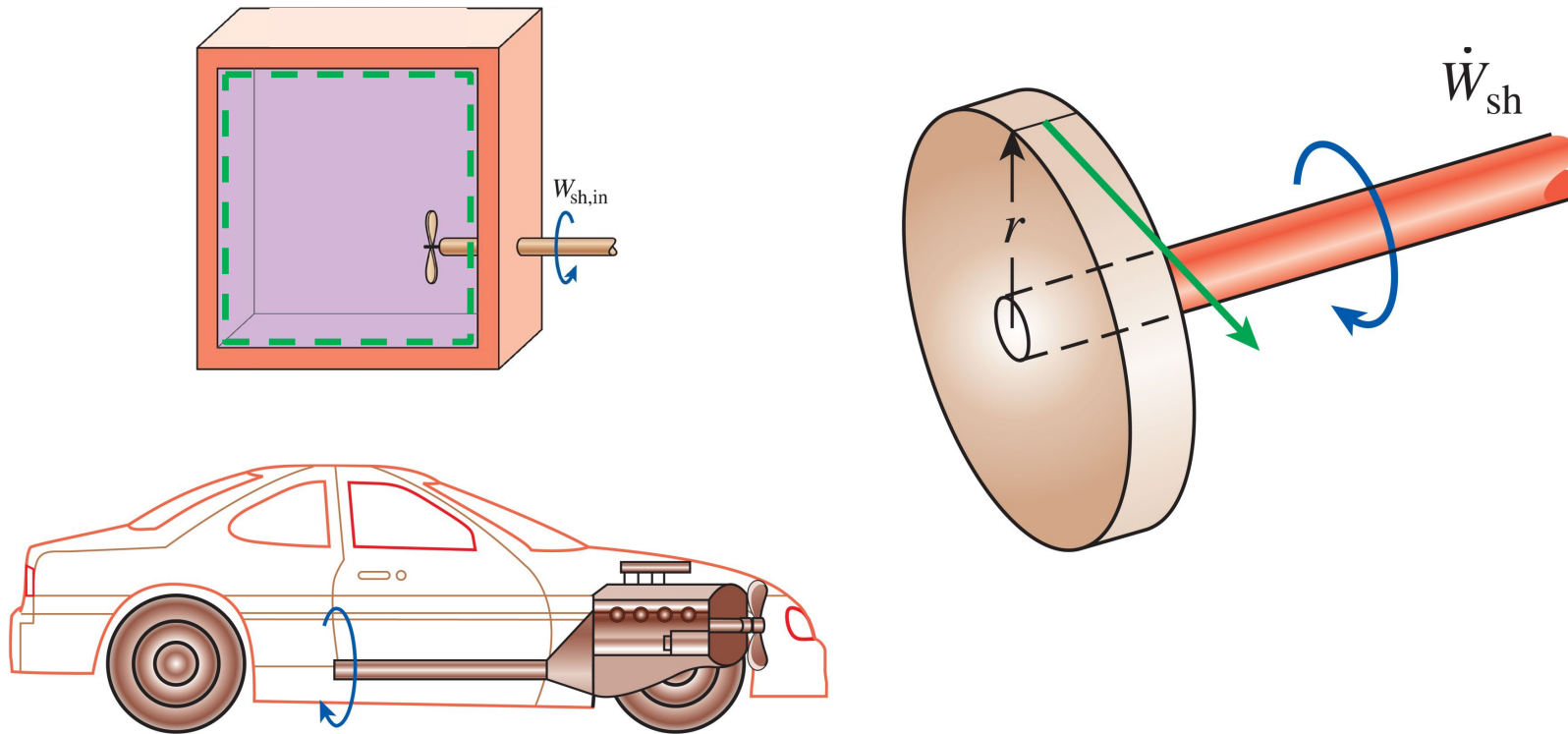
$\dot{W}$ = electrical power [W]
 dt = differential of time [s]
 t_1 = time at state 1 [s]
 t_2 = time at state 2 [s]



Energy transfer by work (3)

In many thermodynamic problems, mechanical work is the only form of work involved. Some common forms of mechanical work are:

- spring work
- work done to raise or to accelerate a body
- shaft work.



Energy transfer by work (4)

The work done during a process between two states (states 1 and 2) is denoted by W_{12} . Work done *per unit mass* (w_{12}) can be calculated as:

$$w_{12} = \frac{W_{12}}{m} \left[\frac{\text{kJ}}{\text{kg}} \right]$$

w = specific work [kJ]
 W = work [kJ]
 m = mass [kg]

The work done *per unit time* is called **power** and is denoted $\dot{W}_{12}$. The unit of power is kJ/s or kW.

Short practice 2

A system receives **30 kJ of work** over a period of **5 seconds**, and has a mass of **2 kg**.

- (a) What is the **power input** $\dot{W}$ to the system?
- (b) What is the **specific work input** w in kJ/kg?
- (c) What does the specific work value represent?

(a) Power Input

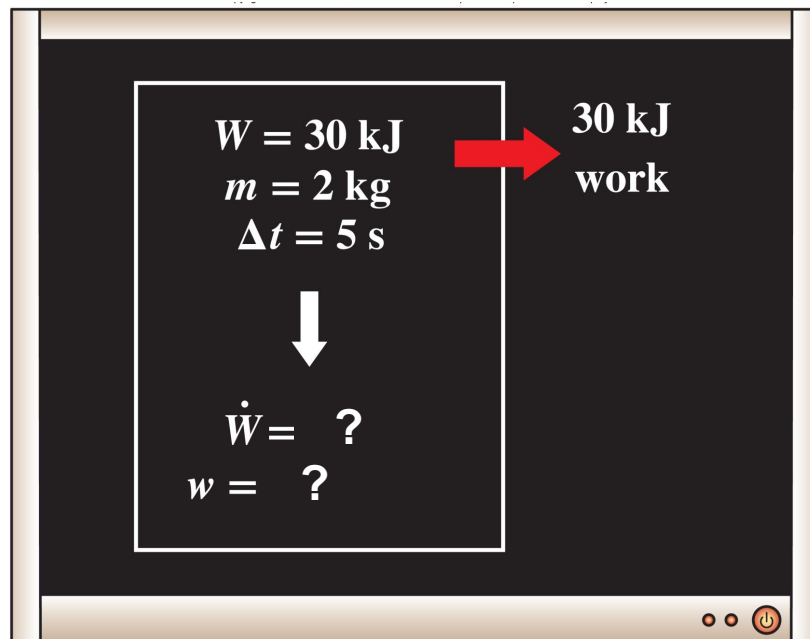
- A. 3 kW
- B. 5 kW
- C. 6 kW
- D. 15 Kw

(b) Specific Work Input

- A. 7.5 kJ/kg
- B. 10 kJ/kg
- C. 15 kJ/kg
- D. 30 kJ/kg

(c) The specific work input represents:

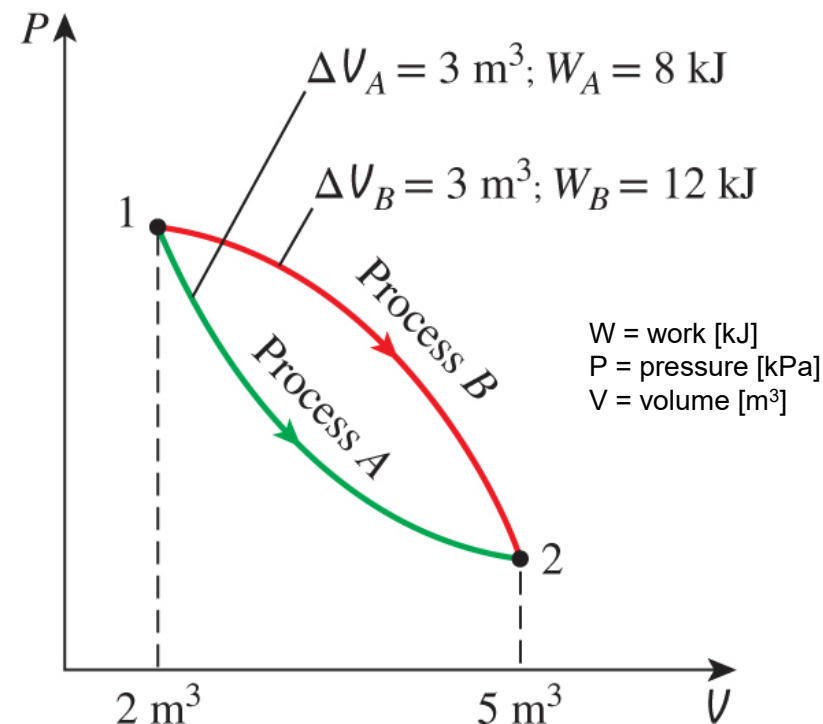
- A. The work done per second.
- B. The total work done over the process.
- C. The work required to raise 1 kg of mass to 1 m height.
- D. The work transferred per unit mass of the system.



Energy transfer by work (5)

Heat vs. Work

- Both are recognized at the boundaries of a system as they cross the **boundaries**
- Systems possess energy, but not heat or work
- Both are associated with a **process**, not a state
- Unlike properties, heat or work has no meaning at a state
- Both are **path functions** (i.e. their magnitudes depend on the path followed during a process as well as the end states)



Properties are point functions have exact differentials (**d**) $\int_1^2 dV = V_2 - V_1 = \Delta V$

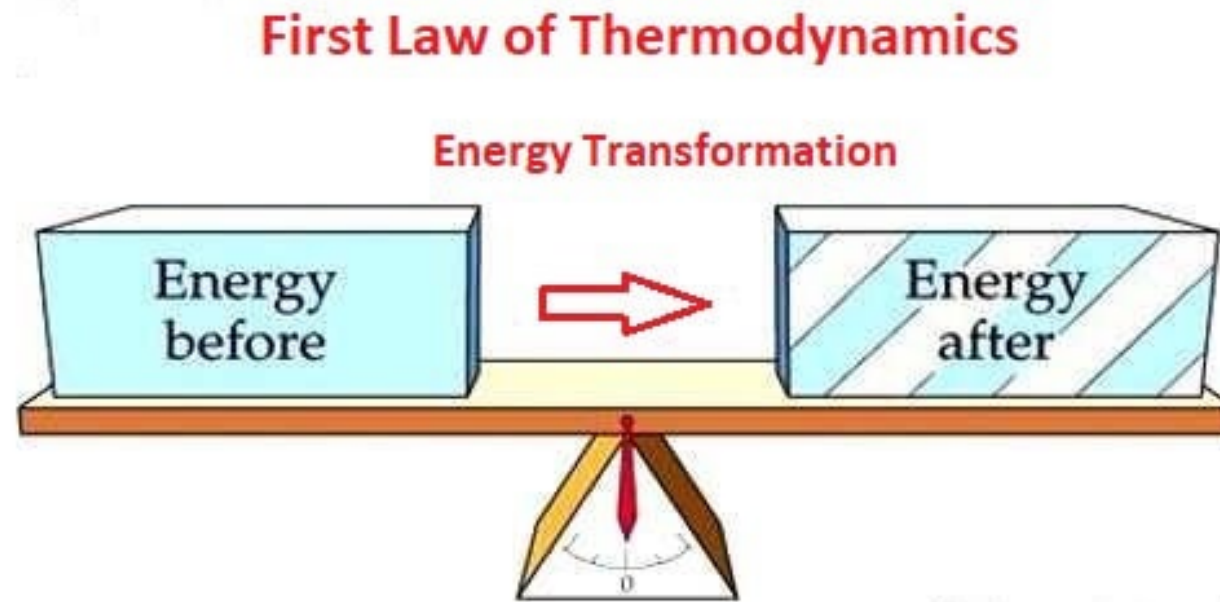
V = volume [m³]

Path functions have inexact differentials (**δ**) $\int_1^2 \delta W = W_{12} \quad (\text{not } \Delta W)$

W = work [kJ]

1st Law of Thermodynamics (1)

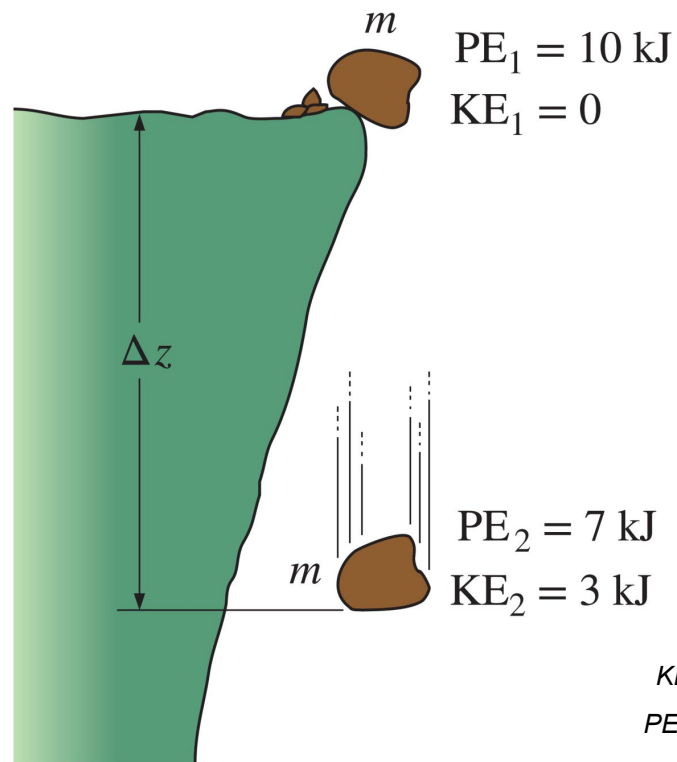
Energy conservation



1st Law of Thermodynamics (2)

1st Law of Thermodynamics

“Based on experimental observations, the first law of thermodynamics states that energy can be neither created nor destroyed during a process; it can only change forms.”

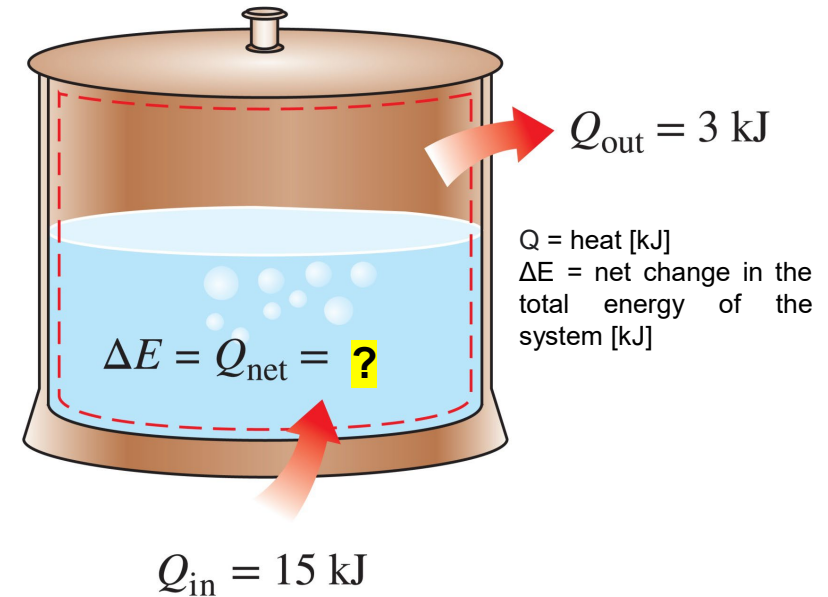
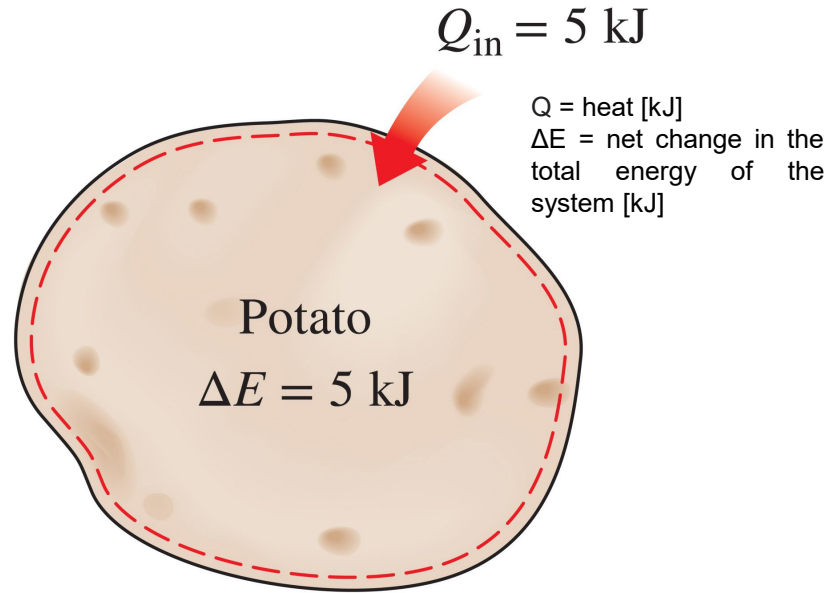


The **1st Law of Thermodynamics**, also known as the **conservation of energy principle**, provides a sound basis for studying the relationships among the various forms of energy and energy interactions.

KE = kinetic energy
 PE = potential energy

1st Law of Thermodynamics (3)

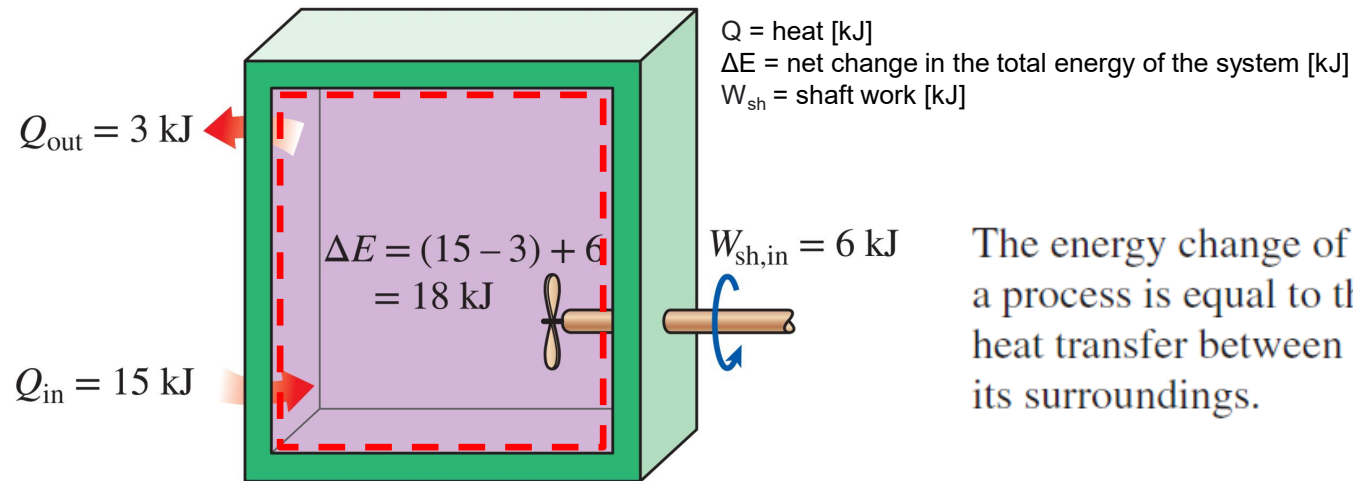
A major consequence of the first law is the existence and the definition of the property *total energy E*.



In the absence of any work interactions, the energy change of a system is equal to the net heat transfer.

1st Law of Thermodynamics (4)

Energy balance



The energy change of a system during a process is equal to the *net* work and heat transfer between the system and its surroundings.

“The net change (increase or decrease) in the total energy of the system during a process is equal to the difference between the total energy entering and the total energy leaving the system during that process.”

$$\underbrace{\left(\text{Total energy entering the system} \right) - \left(\text{Total energy leaving the system} \right)}_{\text{Net energy transfer by heat, work and mass}} = \underbrace{\left(\text{Change in the total energy of the system} \right)}_{\text{Change in internal, kinetic, potential, etc., energies}}$$
$$E_{in} - E_{out} = \Delta E_{system}$$

1st Law of Thermodynamics (5)

Energy change of a system (ΔE_{system}) (for simple compressible systems)

Energy change = Energy at final state – Energy at initial state

$$\underbrace{\Delta E_{\text{system}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work and} \\ \text{mass}}} = \underbrace{E_{\text{final}} - E_{\text{initial}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc., energies}}} = E_2 - E_1$$
$$\Delta E = \Delta U + \Delta KE + \Delta PE$$

$$\text{Internal energy change } \Delta U = m(u_2 - u_1)$$

$$\text{Kinetic energy change } \Delta KE = \frac{1}{2} m(V_2^2 - V_1^2)$$

$$\text{Potential energy change } \Delta PE = mg(z_2 - z_1)$$

E = energy [kJ]

ΔE = net change in the total energy of the system [kJ]

ΔE = net change in the total energy of the system [kJ]

ΔU = internal energy variation [kJ]

ΔKE = kinetic energy variation [kJ]

ΔPE = potential energy variation [kJ]

m = mass [kg]

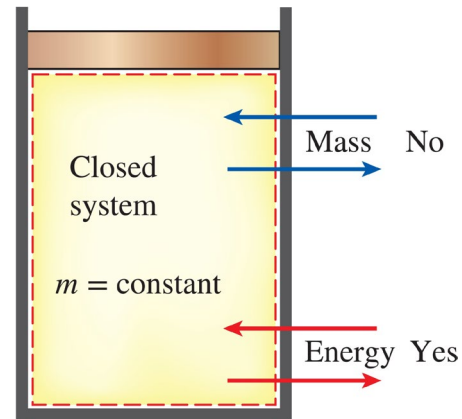
v = velocity of the system relative to some fixed reference frame [m/s]

g = gravitational acceleration [m/s²]

z = elevation of the center of gravity of a system relative to some arbitrarily selected reference level [m]

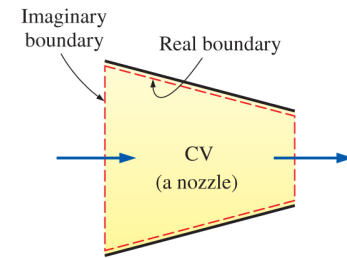
1st Law of Thermodynamics (6)

Mechanisms of energy transfer (E_{in} and E_{out})

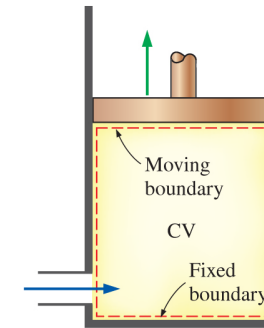


CLOSED SYSTEM

Energy transferred by heat transfer and work transfer



(a) A control volume (CV) with real and imaginary boundaries



(b) A control volume (CV) with fixed and moving boundaries as well as real and imaginary boundaries

OPEN SYSTEM

Energy transferred by heat transfer, mass flow and work transfer

$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) + (E_{mass,in} - E_{mass,out}) = \Delta E_{system}$$

E = energy [kJ]

Q = heat [kJ]

W = work [kJ]

E_{mass} = energy of flowing fluid [kJ]

ΔE = net change in the total energy of the system [kJ]

1st Law of Thermodynamics (7)

Mechanisms of energy transfer (E_{in} and E_{out})

Rate form

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{\text{system}}/dt}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} \quad (\text{kW})$$

For constant rates, the total quantities during a time interval Δt are related to the quantities per unit time as

$$Q = \dot{Q} \Delta t, \quad W = \dot{W} \Delta t, \quad \text{and} \quad \Delta E = (dE/dt) \Delta t \quad (\text{kJ})$$

The energy balance can be expressed on a **per unit mass** basis as

$$e_{in} - e_{out} = \Delta e_{\text{system}} \quad (\text{kJ/kg})$$

Energy balance can also be expressed in the differential form as

$$\delta E_{in} - \delta E_{out} = dE_{\text{system}} \quad \text{or} \quad \delta e_{in} - \delta e_{out} = de_{\text{system}}$$

Summary (1)

	Closed systems	Unit of measurement
Energy balance	$\Delta E = E_2 - E_1 = (Q_{in} - Q_{out}) + (W_{in} - W_{out})$	J
Specific energy balance	$\Delta e = e_2 - e_1 = (q_{in} - q_{out}) + (w_{in} - w_{out})$	J/kg
Energy rate balance	$\frac{dE}{dt} = \dot{E}_2 - \dot{E}_1 = (\dot{Q}_{in} - \dot{Q}_{out}) + (\dot{W}_{in} - \dot{W}_{out})$	W

E = energy [kJ]

Q = heat [kJ]

W = work [kJ]

ΔE = net change in the total energy of the system [kJ]

e = specific energy [kJ/kg]

q = specific heat [kJ/kg]

w = specific work [kJ/kg]

Δe = net change in the total specific energy of the system [kJ/kg]

$\dot{E}$ = energy rate [kW]

$\dot{Q}$ = heat transfer rate [kW]

$\dot{W}$ = power [kW]

dE/dt = rate of change in the total energy of the system [kW]

$$\Delta E = E_2 - E_1 = (U_2 - U_1) + m \cdot \left(\frac{v_2^2}{2} - \frac{v_1^2}{2} \right) + m \cdot g \cdot (z_2 - z_1)$$

Summary (2)

	Closed systems	Unit of measurement
Energy balance	$\Delta E = E_2 - E_1 = (Q_{in} - Q_{out}) + (W_{in} - W_{out})$	J
Specific energy balance	$\Delta e = e_2 - e_1 = (q_{in} - q_{out}) + (w_{in} - w_{out})$	J/kg
Energy rate balance	$\frac{dE}{dt} = \dot{E}_2 - \dot{E}_1 = (\dot{Q}_{in} - \dot{Q}_{out}) + (\dot{W}_{in} - \dot{W}_{out})$	W

E = energy [kJ]

Q = heat [kJ]

W = work [kJ]

ΔE = net change in the total energy of the system [kJ]

e = specific energy [kJ/kg]

q = specific heat [kJ/kg]

w = specific work [kJ/kg]

Δe = net change in the total specific energy of the system [kJ/kg]

$\dot{E}$ = energy rate [kW]

$\dot{Q}$ = heat transfer rate [kW]

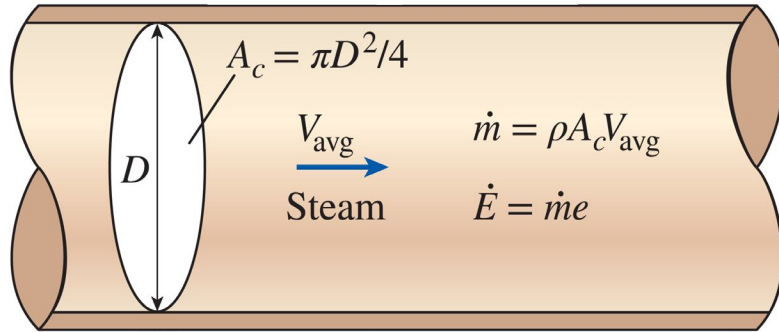
$\dot{W}$ = power [kW]

dE/dt = rate of change in the total energy of the system [kW]

$$\Delta E = E_2 - E_1 = (U_2 - U_1) + m \cdot \left(\frac{v_2^2}{2} - \frac{v_1^2}{2} \right) + m \cdot g \cdot (z_2 - z_1)$$

Summary (3)

Control volumes typically involve fluid flow for long periods of time, and it is convenient to express the energy flow associated with a fluid stream in the rate form.



Mass and energy flow rates associated with the flow of steam in a pipe of inner diameter D with an average velocity of V_{avg} .

volume flow rate = amount of volume flowing through a cross section per unit time (m^3/s)

Mass flow rate:

$$\dot{m} = \rho \dot{V} = \rho A_c V_{avg} \quad (\text{kg/s})$$

mass flow rate = amount of mass flowing through a cross section per unit time (kg/s)

Energy flow rate:

$$\dot{E} = \dot{m}e \quad (\text{kJ/s or kW})$$

$\dot{m}$ = mass flow rate [kg/s]
 ρ = density [kg/m^3]
 $\dot{V}$ = volume flow rate [m^3/s]
 A_c = cross section [m^2]
 V_{avg} = average velocity [m/s]
 e = specific energy [kJ/kg]
 $\dot{E}$ = energy rate [kW]

The Ideal-Gas Equation of State (1)

Any equation that relates some properties (e.g. pressure, temperature and specific volume) of a substance at equilibrium states is called an **equation of state**.

The simplest and best-known equation of state for substances in the gas phase is the ideal-gas equation of state. An ideal gas is an *imaginary* substance that obeys the Ideal-Gas Equation of State (experimentally determined).

$$Pv = RT \quad \text{Ideal-Gas Equation of State}$$

$$R = \frac{R_u}{M} \quad (\text{kJ/kg}\cdot\text{K or kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})$$

R = gas constant

M = molar mass (kg/kmol)

R_u = universal gas constant

P = absolute pressure (kPa)

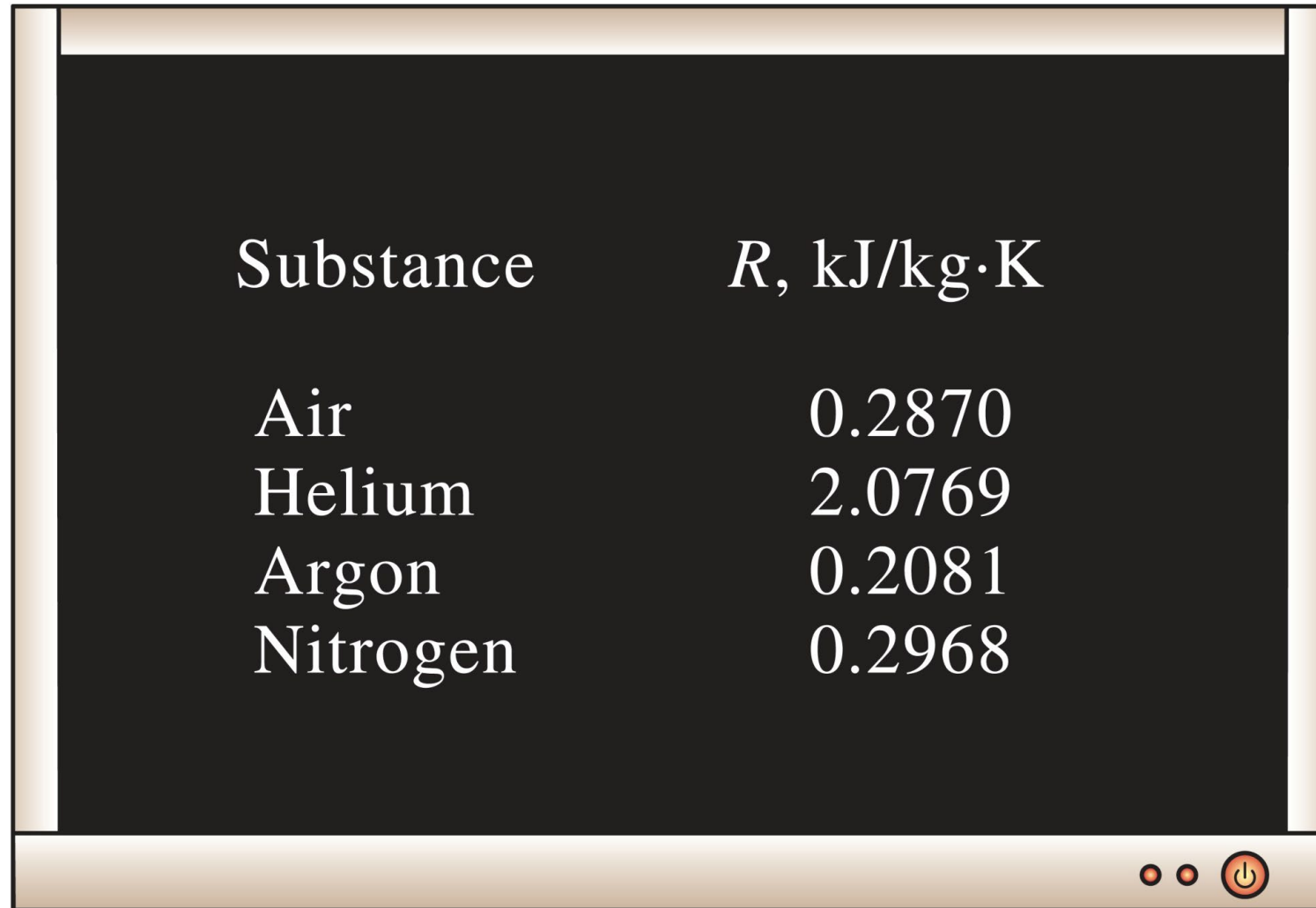
v = specific volume (m^3/kg)

R = gas constant ($\text{kPa}\cdot\text{m}^3/(\text{kg}\cdot\text{K})$)

T = temperature (K)

$$R_u = \begin{cases} 8.31447 \text{ kJ/kmol}\cdot\text{K} \\ 8.31447 \text{ kPa}\cdot\text{m}^3/\text{kmol}\cdot\text{K} \\ 0.0831447 \text{ bar}\cdot\text{m}^3/\text{kmol}\cdot\text{K} \\ 1.98588 \text{ Btu/lbmol}\cdot\text{R} \\ 10.7316 \text{ psia}\cdot\text{ft}^3/\text{lbmol}\cdot\text{R} \\ 1545.37 \text{ ft}\cdot\text{lbf/lbmol}\cdot\text{R} \end{cases}$$

The Ideal-Gas Equation of State (2)



Substance	R , kJ/kg·K
Air	0.2870
Helium	2.0769
Argon	0.2081
Nitrogen	0.2968

The Ideal-Gas Equation of State (3)

Various expressions of ideal gas equation

$$V = m\upsilon \longrightarrow PV = mRT$$

V = volume (m^3)

m = mass (kg)

υ = specific volume (m^3/kg)

P = absolute pressure (kPa)

V = volume (m^3)

m = mass (kg)

R = gas constant ($kPa \cdot m^3/(kg \cdot K)$)

T = temperature (K)

Mass

$$mR = (\underbrace{MN})R = NR_u \longrightarrow P\upsilon = NR_uT$$

Molar mass (M) x Mole number (N)

$$V = N\bar{\upsilon} \longrightarrow P\bar{\upsilon} = R_uT$$

Molar specific volume (m^3/kg)

$$\frac{P_1\upsilon_1}{T_1} = \frac{P_2\upsilon_2}{T_2}$$

Ideal gas equation at two states for a fixed mass

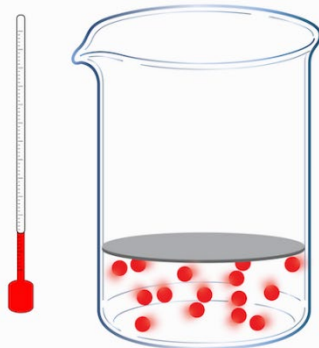
Per unit mass	Per unit mole
υ , m^3/kg	$\bar{\upsilon}$, $m^3/kmol$
u , kJ/kg	$\bar{u}$, kJ/kmol
h , kJ/kg	$\bar{h}$, kJ/kmol

The Ideal-Gas Equation of State (4)

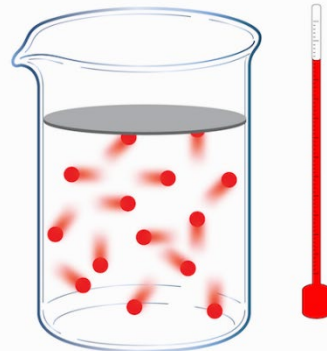
A gas is considered ideal if:

- there are great distances between molecules (low pressure, i.e. low density)
- there are negligible attractive forces between molecules (high temperature, i.e. low density).

Low temperature



High temperature



(a) Low pressure



(b) High pressure

The Ideal-Gas Equation of State (5)

- At pressures below 10 kPa, water vapour can be treated as an ideal gas
- In air-conditioning applications, the water vapour in the air can be treated as an ideal gas (as its pressure is very low)
- Dense gases, such as water vapor in steam power plants (see Rankine cycles in your textbook) and refrigerant vapor in refrigerators and heat pumps (see Refrigeration cycles in your textbook), however, should not be treated as ideal gases.